

HomeSense

An IoT-Based Smart Home Organization and Assistance System for Independent Living of Visually Impaired Individuals.

Submitted for

DEVS REC Board Recruitment 2026 – Round 2.

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Abstract:

Visually impaired individuals often depend on family members or caregivers to locate everyday household items such as medicines, groceries, important documents, and kitchen essentials. While existing assistive technologies mainly focus on navigation and obstacle detection, there is limited emphasis on improving independent living within home environments.

HomeSense is an IoT-based smart home organization and assistance system designed to help visually impaired individuals independently locate frequently used household items through voice-guided interaction. The system combines RFID-tagged objects, ESP32-based smart controllers, and a mobile application to maintain a digital inventory of item locations. Instead of relying on expensive wearable devices or continuous camera usage, HomeSense creates an intelligent home environment that remembers where essential items are stored and provides private audio guidance whenever needed.

The proposed solution focuses on four key principles: independence, privacy, dignity, and affordability. Although presented as a conceptual design, the architecture is practical, scalable, and suitable for future implementation in smart homes.

Keywords: Internet of Things (IoT), ESP32, RFID, Smart Home, Accessibility, Visual Impairment, Independent Living.

1. Introduction:

Many of us go through life taking our abilities for granted because we have never had to imagine life without them. Yet people who face physical challenges every day often teach us the greatest lessons in resilience, determination, and courage. They remind us to value what we have and to use it wisely. If technology can make even one part of their daily life easier, then even the smallest innovation can create a meaningful difference.

Visually impaired individuals often rely on family members to locate medicines, grocery items, kitchen containers, important documents, and other household essentials. While navigation technologies have significantly improved outdoor mobility, independent living inside the home remains a challenge for many.

HomeSense proposes a different approach by making the **home environment itself intelligent**. Instead of expecting users to remember the location of every object or depend on caregivers, the proposed IoT-based system maintains a digital inventory of tagged household items and provides private voice-based guidance whenever required.

The objective of HomeSense is not to replace human assistance but to reduce unnecessary dependence in daily activities while preserving privacy, dignity, and affordability.

2. Problem Statement

Visually impaired individuals often depend on caregivers or family members to locate household items such as medicines, groceries, documents, clothing, and kitchen essentials. Existing assistive technologies primarily focus on outdoor navigation, obstacle detection, or object recognition, while everyday organization inside the home remains largely dependent on human assistance. This dependence can reduce confidence and limit independent living. Therefore, there is a need for an affordable, privacy-preserving IoT solution that enables users to organize, remember, and locate household items independently through simple voice interaction.

3. Objectives

- To develop an IoT-based smart home organization system for visually impaired individuals.
- To enable voice-assisted searching of household items.
- To store and retrieve item locations using RFID technology.
- To reduce dependence on caregivers for daily household tasks.
- To provide a privacy-preserving and affordable solution.
- To design a scalable system suitable for future smart home applications.

4. Existing Solutions

Current assistive technologies include:

Smart Canes

- Help detect nearby obstacles.
- Limited to navigation.
- Cannot identify or locate household items.

AI Camera Applications

- Recognize objects using smartphone cameras.
- Require proper lighting and camera positioning.
- Often depend on internet connectivity.

RFID-Based Object Finders

- Help identify tagged objects.
- Mainly designed for individual item tracking.
- Lack integrated home organization and memory assistance.

Voice Assistants

- Answer questions and set reminders.
- Cannot accurately maintain the physical locations of household items.

5. Limitations of Existing Systems

- High implementation cost.
- Dependence on continuous internet connectivity.
- Privacy concerns due to continuous camera usage.
- Limited support for organizing household environments.
- No integrated memory assistance for remembering item locations.
- Focus mainly on navigation rather than independent living inside homes.

6. Proposed Solution

HomeSense

HomeSense is an IoT-based smart home organization and assistance system that enables visually impaired individuals to independently locate and manage household items.

The proposed system combines RFID technology, ESP32 microcontrollers, voice interaction, and a mobile application to create an intelligent home environment. Each frequently used object is attached with an RFID tag, while RFID readers connected to ESP32 nodes monitor designated storage locations such as shelves, drawers, cabinets, and medicine boxes.

When a user asks,

"Where is my medicine?"

the application retrieves the stored location and privately provides voice guidance such as

"Bedroom cabinet, top shelf."

Additionally, HomeSense includes a **Smart Memory Mode**, allowing users to save custom locations using voice commands.

Example:

"Remember that I kept my passport in the second drawer."

The system stores this information and retrieves it whenever requested.

Unlike many existing solutions, HomeSense emphasizes affordability, privacy, and independent living by avoiding continuous camera monitoring while utilizing low-cost IoT components.

7. System Architecture

Architecture Description

The HomeSense system consists of three major layers:

1. User Layer

- Android Mobile Application
- Voice Input (Speech-to-Text)
- Voice Output (Text-to-Speech)

2. IoT Layer

- ESP32 Controller
- RFID Reader (RC522)
- RFID Tagged Household Objects
- Wi-Fi Communication

3. Cloud Layer

- Firebase Database
- User Preferences
- Item Location Database
- Smart Memory Records

The mobile application communicates with the ESP32 through Wi-Fi to retrieve item locations stored in the database. Users interact with the system using simple voice commands, ensuring a hands-free and accessible experience.

8. Hardware Components

Component	Purpose
ESP32	Main microcontroller that manages communication between RFID reader, mobile application, and database.
RFID Reader (RC522)	Reads RFID tags attached to household items.
RFID Tags	Uniquely identify important household objects.
Wi-Fi Module (Built into ESP32)	Enables communication between hardware and mobile application.
Smartphone	Provides voice interaction and audio guidance.
Power Adapter	Supplies power to ESP32 nodes.

9. Software Components

Software	Purpose
Flutter	Mobile application development
Firebase	Cloud database
Arduino IDE	Programming ESP32
Speech-to-Text	Converts voice commands into text
Text-to-Speech	Provides private audio guidance

10. Working Methodology

Step 1

The user attaches RFID tags to frequently used household items.

Step 2

The corresponding locations of these tagged items are stored in the HomeSense database.

Step 3

When the user asks a voice query such as:

"Where is my medicine?"

the mobile application converts speech into text.

Step 4

The ESP32 retrieves the corresponding location information from the database.

Step 5

The application privately provides voice guidance such as:

"Medicine Box – Bedroom Cabinet – Top Shelf."

Step 6

If the user changes the location of an item, the Smart Memory Mode updates the stored information for future retrieval.

11. Advantages

- Promotes independent living.
- Preserves user privacy.
- Affordable implementation using low-cost IoT hardware.
- Reduces dependence on caregivers.
- Simple voice-based interaction.

- Easily scalable for larger smart home environments.
- Supports future integration with additional smart home devices.

12. Applications

- Smart Homes
- Assisted Living Facilities
- Elderly Care Homes
- Rehabilitation Centers
- Independent Living Programs
- Smart Residential Communities

13. Future Scope

Future enhancements may include:

- AI-based object recognition.
- Medicine reminder and dosage management.
- Integration with smart refrigerators.
- Indoor navigation assistance within homes.
- Automatic detection of misplaced items.
- Multi-language voice support.
- Integration with wearable assistive devices.

14. Conclusion

HomeSense presents a practical IoT-based approach to improving independent living for visually impaired individuals by enabling efficient organization and retrieval of household

items. Rather than replacing human assistance, the proposed system reduces unnecessary dependence through affordable technology, voice interaction, and smart home organization. By emphasizing privacy, dignity, and accessibility, HomeSense demonstrates how existing IoT technologies can be integrated into a meaningful solution that enhances everyday life. The proposed architecture also provides a scalable foundation for future smart home innovations.